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TerminalTotaltoken.py x

```
1 import os
2 import openai
3 import secret
4 openai.api_key='sk-QVkkVGvk7ose2G3HCQM7T3B1bkFJI8YuDKIMdS8Rk6rTsg1Y'
5
6
7
8
9
10 message= [{"role": "system", "content": "You are a helpful assistant."},
11 {"role": "user", "content": "Tell me about the history of the Eiffel Tower and its significance."}]
12
13
14
15 ## put yout code above this line
16 response=openai.ChatCompletion.create(
17     model="gpt-3.5-turbo",
18     messages=message
19 )
20 print(response['usage']['total_tokens'])
```

Guide x

Collapse

<>⚙️☰

Managing Conversation Length and Token Limits

We will explore the importance of managing conversation length, understanding token limits, and techniques for optimizing conversations to ensure informative and better responses. Token limits and conversation length play a significant role in the quality and completeness of responses generated by the **ChatGPT API**.

Language models read text in chunks called tokens. In the ChatGPT as Language models, also uses tokens to represent text. In English, a token can be as short as one character or as long as one word (e.g., a or apple), and in some languages tokens can be even shorter than one character or even longer than one word. For example, the string `ChatGPT is great!` is encoded into six tokens: `["Chat", "G", "PT", " is", " great", "!"]`.

Both input and output tokens count toward the total tokens used in an API call. Each model has a maximum token limit (e.g., 4096 tokens for gpt-3.5-turbo). If a conversation exceeds this limit, you must truncate, omit, or shorten parts of the text to fit within the constraints. This can impact the coherence and completeness of the generated responses.

Techniques for Optimizing Conversations:

1. **Prioritize important information:** Keep the most relevant parts of the conversation while removing or shortening less essential content.

2. **Condense messages:** Use concise language and remove unnecessary words or phrases in user messages.

3. **Remove excessively long responses:** If a previous AI-generated response is too long and not crucial for the current query, consider removing or shortening it.

Be aware of the token count: Monitor the total tokens used in a conversation to ensure that you remain within the model's token limit.

For example, suppose you have the following conversation that exceeds the token limit:

```
[{"role": "system", "content": "You are a helpful assistant."}, {"role": "user", "content": "Tell me about the history of the Eiffel Tower and its"}, {"role": "assistant", "content": "<A long, detailed response about the Eiffel Tower"}, {"role": "user", "content": "What materials were used to build the Eiffel Tower and"}]
```

TRY IT!

Terminated by timeout

To see how many tokens are used by an API call, check the usage field in the API response:

```
response['usage']['total_tokens']
```

TRY IT!

Replace the message with this Optimized Conversation:

```
[ {"role": "system", "content": "You are a helpful assistant."}, {"role": "user", "content": "Tell me about the history of the Eiffel Tower and its"}, {"role": "assistant", "content": "<A shortened, concise response about the Eiffel Tower"}, {"role": "user", "content": "Materials used and construction process?"}]
```

TRY IT!

In the optimized conversation, we shortened the user messages and the assistant's response to fit within the token limit while still providing useful information.

Every GPT response will include a `finish_reason`. Which of the following reason could be generated if its due to reaching token limit?

☐ stop

☒ length

☐ content_filter

☐ null

`length`: Incomplete model output due to `max_tokens` parameter or token limit

Check It!

Next

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https://codio.com/sandipan-dey/enhancing-chatgpt-api-interactions/tree/Totaltoken.py

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